

NEET Important Chapter – Alternating Current

Alternating Current: An Important Concept for NEET

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The chapter of Alternating Current starts with the basic revision of all the concepts we have studied in our lower grades and helps to build a connection between the advanced parts of Alternating Current. In the following chapter, we have discussed alternating and direct current, average and rms value of alternating current, resistance offered by various elements (inductor, resistor and capacitor) to a.c , impedances and phases of ac circuits containing different elements, etc.

While deriving the formulae and equations for resistance offered by various elements (inductor, resistor and capacitor) to a.c., L.C Oscillations, we will also study about transformers and the power losses in a transformer with examples in the chapter Alternating Current.

In this chapter, students will also get to learn about the Power in AC Circuit and Resonance, where we will study about the sharpness of resonance along with other major topics.

Now, let us move on to the important concepts and formulae related to NEET exams along with a few solved examples.

Important Topics of Alternating Current

- Alternating Current
- Impedance
- Average and RMS value of AC
- Admittance
- Reactance
- Resonance
- Power Factor
- L. C. Oscillations
- Transformer

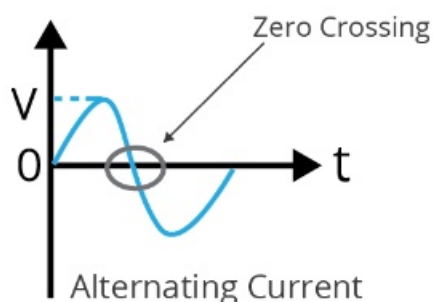
Important Concepts of Alternating Current

Sl.No	Name of the Concept	Key Points of the Concept
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1.

Alternating Current and Direct Current

- If a current periodically changes its direction and magnitude then it is termed as alternating current. The change can be seen as it first increases from zero to a maximum value, then decreases to zero and reverses in direction, increases to a maximum in this direction and then decreases to zero.



- On the other hand, direct current is that which can change the magnitude but never changes its direction.

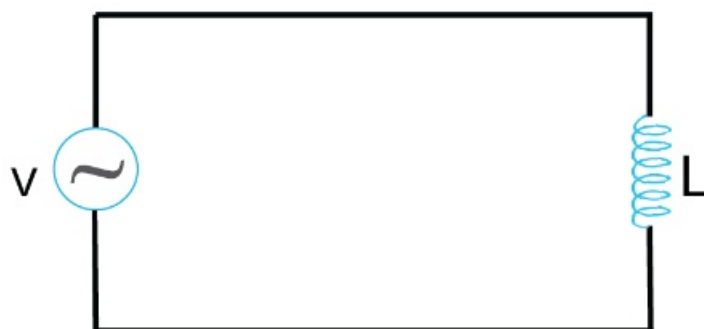
(image will be uploaded soon)

2.	Average and RMS value of current	• The average value for an AC cycle is zero because of equal positive and negative cycles. • The rms value of AC is defined as the direct current producing the same heating effect in a given resistor in a given time as is produced by the given A.C. through the same resistor for the same time period. • $I_{rms} = \frac{I_m}{\sqrt{2}}$
3.	AC Circuit Containing Resistance Only	• For a pure resistive circuit where only the resistor is connected in the circuit with some alternating voltage source and current • Current and voltage here are in phase. (image will be uploaded soon) • Impedance of the circuit here is the resistance only.

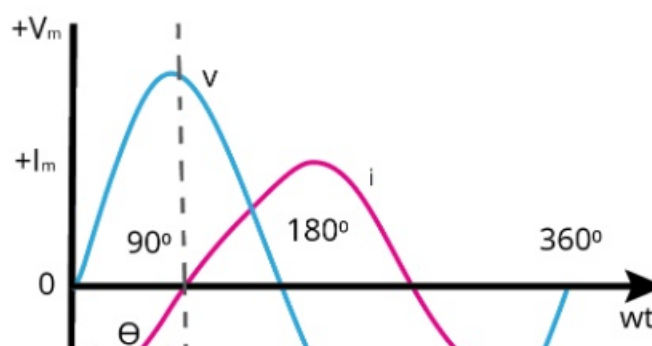
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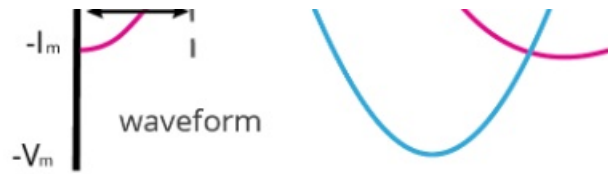
AC Circuit Containing Inductor Only

- For a pure inductive circuit where only the inductor L is connected in the circuit with some alternating voltage source, the current flowing through the inductor circuit is —



- Current in this circuit lags behind the applied voltage source by phase difference of —

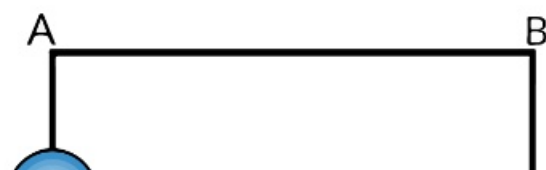




- Impedance of the circuit is called inductive reactance = ωL

- For a pure capacitive circuit where only the capacitor C is connected in the circuit with some alternating voltage source, the current flowing through the capacitor circuit is

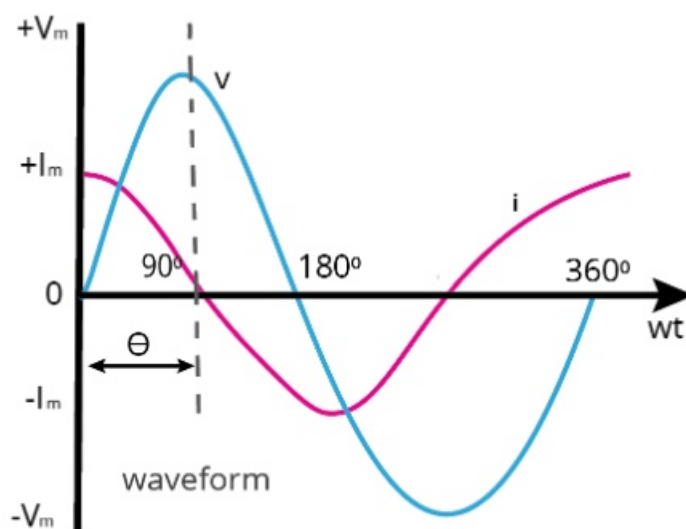
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5. AC Circuit Containing Capacitor Only



- Current in this circuit leads the applied voltage source by phase difference of —



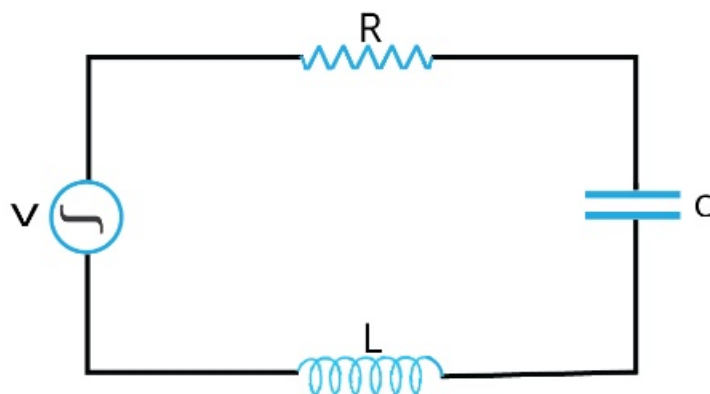
- Impedance of the circuit is called capacitive reactance $= 1/\omega C$

6.

AC Circuit Containing Resistor, Inductor and Capacitor in series

- For a series circuit where resistor, inductor and capacitor are connected in series in the circuit with some alternating voltage source

the current flowing through the series circuit is



- Here the impedance of the circuit is

- And Phase = _____

7.

Resonance

- Resonance is one of the main characteristics of an LCR circuit which means that if a particular frequency matches with the natural frequency of the system then that condition is called

		resonance. • $\frac{1}{Z} = \frac{1}{R}$ this is called resonant frequency.
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8.	Power in AC Circuit	• In AC circuit, power for one cycle is defined as the rate of doing work hence if • $P = VI \cos \phi$ • Current = $I = \frac{V}{Z}$ • Then power = $P = VI \cos \phi$
9.	Transformer	• A device which is used for converting either high voltage to low voltage or vice versa is called a transformer. • A transformer is classified into two categories: • Step-up Transformer • Step-down Transformer

Sl. No	Name of the Concept	Formula
1.	Rms value of current	$I_{rms} = \frac{I_0}{\sqrt{2}}$
2.	Inductive Reactance	$X_L = 2\pi fL$
3.	Capacitive Reactance	$X_C = \frac{1}{2\pi fC}$
4.	Impedance of LCR Circuit	$Z = \sqrt{R^2 + (X_L - X_C)^2}$
5.	Resonance	$\frac{1}{Z} = \frac{1}{R}$ this is called Resonant frequency. At resonance, Z is minimum and current is maximum i.e. $I_{rms} = \frac{V_{rms}}{R}$
6.	Power in AC Circuit	Power = $V_{rms} I_{rms} \cos \phi$
7.	Efficiency of	$\eta = \frac{P_{out}}{P_{in}} \times 100\%$ Here V_0 , V are the output voltage and

∴	Transformer	current of secondary coil and , are the input voltage and current of the primary coil.
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Solved Examples of Alternating Current

1. The current in an inductive circuit is given by $0.3 \sin (200t - 40^\circ)$ A. Write the equation for the voltage across it if the inductance is 40 mH.

Sol:

Given, $i = 0.3 \sin (200t - 40^\circ)$ A, $L = 40\text{mH} =$

$$= \quad = \quad = 8\Omega$$

$$= \quad = \quad = 2.4\text{V}$$

In an inductive circuit, the voltage leads the current by 90° .

Therefore,

$$V =$$

$$V =$$

$$V = \quad .$$

Key point: The ques here focuses mainly on the phase difference between current and voltage in an inductive circuit.

2. The output of a step-down transformer is measured to be 24 V when connected to a 12 W light bulb. The value of the peak current is

a. $\frac{1}{2}$

b. $\frac{1}{\sqrt{2}}$

c. $\frac{1}{2\sqrt{2}}$

d. $\frac{1}{\sqrt{2}}$

Sol:

Given:

Output of Secondary Voltage = $V_s = 24V$

Power associated with Secondary = $P_s = 12W$

$$I_s = \frac{P_s}{V_s} = \frac{12}{24} = 0.5A$$

Amplitude of the current in the secondary winding:

$$I_m = \frac{I_s}{\frac{1}{\sqrt{2}}}$$

$$I_m = 0.5(1.414) = 0.707 = \frac{1}{\sqrt{2}} A.$$

Hence, Option (a) is Correct.

Key Point: Here, in the case of a step-down transformer, the output voltage decreases and the output current increases.

Previous Year Questions of Alternating Current

1. An inductor 20 mH, a capacitor 50 μ F and a resistor 40 Ω are connected in series across a source of emf $V=10 \sin 340t$. The power loss in A.C. circuit is (NEET - 2016)

a. 0.51 W

b. 0.67 W

c. 0.76 W

d. 0.89 W

Sol:

Given:

$$V=10 \sin 340t = \frac{10}{\sqrt{2}}$$

$$L = 20mH =$$

$$C = 50\mu F =$$

$R = 40\Omega$ and $V = 10 \sin 340t$

$$= \frac{10}{\sqrt{2}} = \frac{10}{1.414} = 7.07 \text{ V}$$

$$= \frac{7.07}{\sqrt{2}} = 5.0 \text{ V}$$

$$= \frac{5.0}{\sqrt{2}} = 3.54 \text{ V}$$

Hence $P =$

Or $P =$

$=$

$=$

$$= 0.47 \text{ W}$$

Correct solution is 0.47W and the closest option is (a).

Trick: Here the formulae of capacitive, inductive reactance and impedance play a major role in ans along with the power.

3. A series LCR circuit is connected to an AC voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{2}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{2}$ between current and voltage. The power factor of the circuit is (NEET 2020)

a. 0.51W

b. 0.67W

c. 0.76W

d. 0.89 W

Sol:

Given:

Given,

When L is removed,

$$\text{Phase} = \quad = \quad = \quad = \quad$$

When C is removed,

$$\text{Phase} = \quad = \quad = \quad = \quad$$

From both above-mentioned equations, we have

Hence, the circuit is in resonance and $Z = R$

So, power factor = $\quad$

The correct answer is 1 and option is b.

Trick: Formula of phase is given as = $\quad$

Practice Questions

1. A 100 mH inductor, a 25 μF capacitor and a 15 Ω resistor are connected in series to a 120 V, 50 Hz a.c. source. Calculate

(a) impedance of the circuit at resonance.

(b) current at resonance.

(c) Resonant frequency.

(Ans (a) 15 Ω (b) 8 A (c) 100.7 Hz)

2. The inductance of a choke-coil is 0.2 Henry and its resistance is 0.50 Ω . If a current of 2.0 ampere (rms value) and frequency 50Hz is passed through it, what will be the potential difference across its ends ?

(Ans: 125.6 V)

Conclusion

In conclusion we can say that the chapter Alternating Current has covered all the important formulae and concepts like Power factor, Impedance, Resistance offered

important formulae and concepts like power factor, impedance, reactance offered by different electrical circuits, etc. which will help in boosting your confidence to score high marks in the [NEET 2022](#) exam. Students can have hands-on experience of the exam by practising the numericals.

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